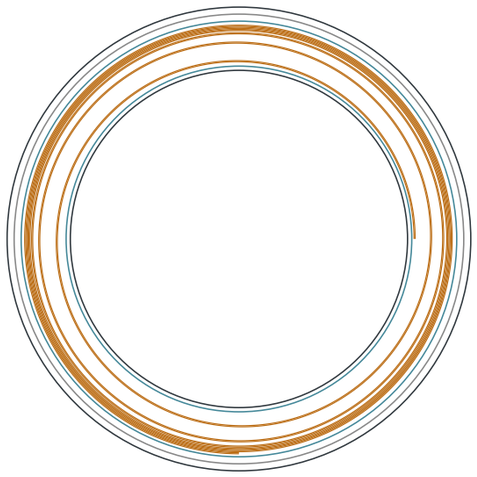
REPAIRED CONSTANT-LENGTH ROTARY COLLAR / GEOMETRIC PROTOTYPE

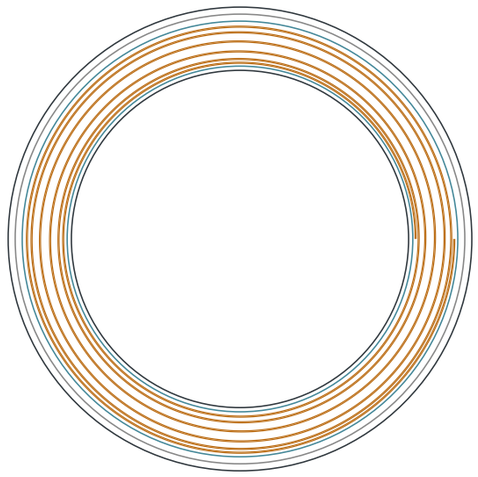
Ø48 × 142 stationary core; Ø66 collar. Supplier, strength, cycle life and source-board integration remain open.

-90 degrees / active ribbon



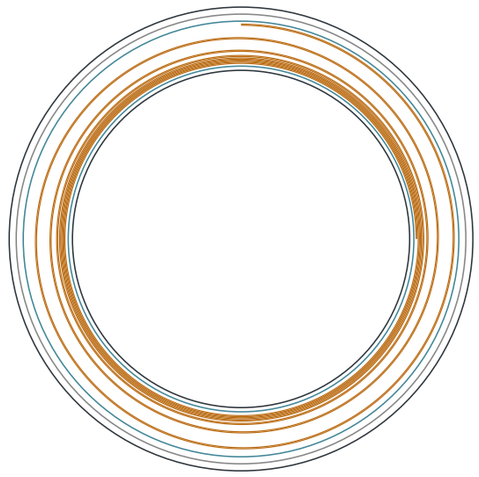
Ø66 / Ø48; 3 drawing units per mm

+0 degrees / active ribbon



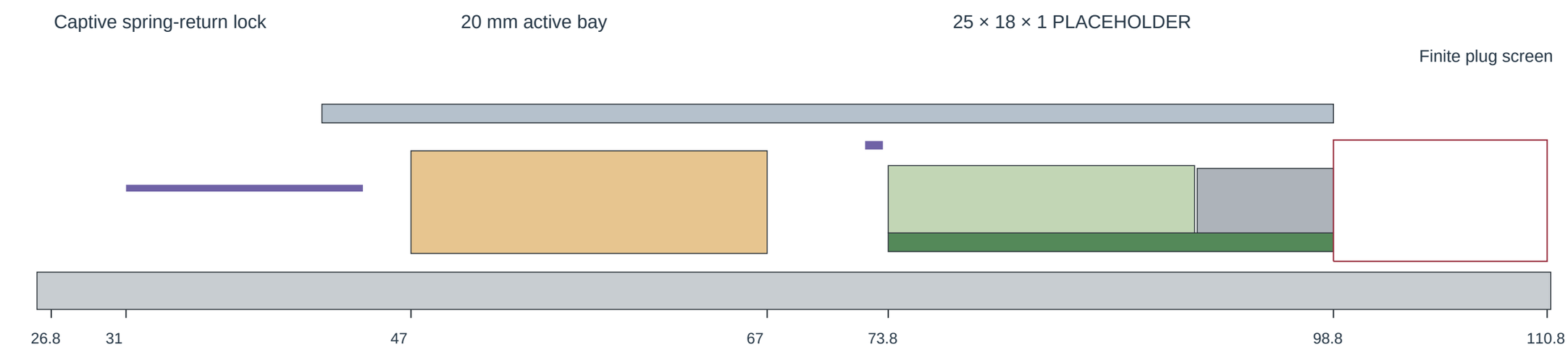
Ø66 / Ø48; 3 drawing units per mm

+90 degrees / active ribbon



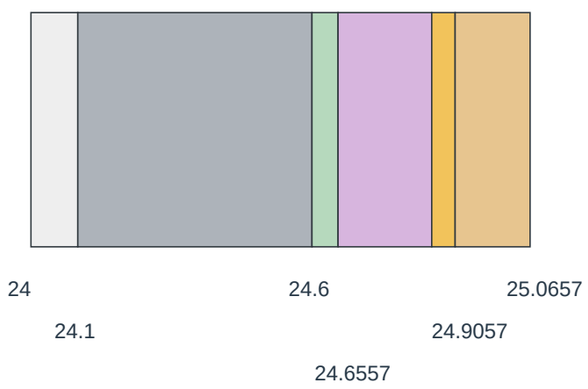
Ø66 / Ø48; 3 drawing units per mm

AXIAL ENVELOPE — lock X31..44.3; active flex X47..67; target board X73.8..98.8



Locked length67.8 mm; release sweep72.0 mm. Axial/radial drawing scales10.5 /11 units per mm; see dimensions.json.

REPAIRED FIXED CLAMP / ENLARGED RADIAL STACK



Backing outerR24.6; stiffener0.20 + adhesive0.05; minimum unpadded gap0.055676 mm.

Contoured compliant ear pads fill that gap outside the active20 mm width.

Stiffener/adhesive taper to zero over0.08 rad (~2.0 /2.44 mm), fixed/moving tails.

Both rotor halves and stationary spool are connected single closed solids.

Four frame head counterbores; radial ear-driver passages; captive service cap.

Power reserve3.46 max height +0.15 assembly allowance; no actual source-PCB fit claim.

Engineering geometry only. Taper process, bond/preload, spring sourcing, flex equilibrium/endurance, thermal and fixture fit require qualification.